CLASS 9 — MATHS · EASY LEVEL UNIT TEST SET 2

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The next term of the GP $1, 2, 4, 8, \ldots$ is: 1

(a) 17 **(b)** 16 **(c)** 15 **(d)** 18

2. An arc subtends 20° at a point on the circle. At the centre it subtends: 1

(a) 30° **(b)** 60° **(c)** 50° **(d)** 40°

3. The area of a parallelogram with base 15 cm and height 8 cm is (cm^2): 1

(a) 120 **(b)** 119 **(c)** 122 **(d)** 121

4. The probability of an impossible event is: 1

(a) 0 **(b)** -1 **(c)** $\frac{1}{6}$ **(d)** $\frac{1}{2}$

5. The common ratio of the GP $3, 6, 12, 24, \ldots$ is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 1

6. $ABCD$ is a cyclic quadrilateral with $\angle A = 65^\circ$. Then $\angle C =$ 1

(a) 105° **(b)** 115° **(c)** 135° **(d)** 125°

7. The circumference of a circle of radius 7 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 43 **(b)** 46 **(c)** 45 **(d)** 44

8. The number of outcomes when a die is rolled once is: 1

(a) 5 **(b)** 6 **(c)** 7 **(d)** 8

9. The sum of the first 6 natural numbers is: 1

(a) 20 **(b)** 21 **(c)** 22 **(d)** 27

10. The longest chord of a circle is its: 1

(a) tangent **(b)** radius **(c)** secant **(d)** diameter

11. The area of a circle of radius 7 cm (take $\pi = \frac{22}{7}$) is (cm^2): 1

(a) 153 **(b)** 154 **(c)** 155 **(d)** 156

12. If $P(E) = \frac{2}{7}$, then $P(\text{not } E)$ is: 1

(a) $\frac{6}{7}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{5}{7}$ **(d)** $\frac{2}{7}$

13. The sequence $2, 4, 6, 8, \ldots$ is: 1

(a) an AP **(b)** a GP

(c) neither an AP nor a GP **(d)** both an AP and a GP

14. AB is a diameter and C is a point on the circle. Then $\angle ACB =$ 1

(a) 120° **(b)** 135° **(c)** 90° **(d)** 60°

15. A circle has circumference 44 cm (take $\pi = \frac{22}{7}$). Its radius is (cm): 1

(a) 6 **(b)** 8 **(c)** 7 **(d)** 9

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the area of a circle of radius 7 cm (take $\pi = \frac{22}{7}$). 2

17. A bag contains 5 red and 3 white balls. Find the probability of drawing a red ball. 2

18. Find the common difference of the AP $6, 10, 14, 18, \ldots$ and write the next two terms. 2

19. AB is a diameter of a circle and C is a point on it with $\angle CAB = 28^\circ$. Find $\angle ABC$. 2

20. Find the area of a triangle with base 18 cm and height 10 cm. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the sum of the first 20 natural numbers. 3

22. $ABCD$ is a cyclic quadrilateral with $\angle A = 70^\circ$ and $\angle B = 85^\circ$. Find $\angle C$ and $\angle D$. 3

23. Find the area of a trapezium with parallel sides 10 cm and 14 cm and height 6 cm. 3

24. A ticket is drawn at random from tickets numbered 1 to 30. Find the probability that it shows a multiple of 5. 3

OR

Find the 6th term of the GP with first term 1 and common ratio 2. 3

25. A chord of a circle of radius 5 cm is at a distance of 4 cm from the centre. Find the length of the chord. 3

OR

The area of a parallelogram is 120 cm^2 and its base is 15 cm. Find its height. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. AB is a diameter of a circle and C, D lie on the circle on opposite sides of AB. If $\angle CAB = 25^\circ$ and $\angle DAB = 30^\circ$, find $\angle ABC$ and $\angle ABD$. 5

OR

A circular park has radius 14 m. Find its circumference and area (take $\pi = \frac{22}{7}$), and the cost of fencing it at ₹15 per metre. 5

27. A fair die is rolled once. Find 5

(i) $P(\text{an even number})$,

(ii) $P(\text{a number greater than } 4)$ and

(iii) $P(\text{an even number or a number greater than } 4)$ using the favourable outcomes.

OR

An AP has first term 2 and common difference 3. Find the sum of its first 10 terms. 5

END OF QUESTION PAPER — Total: 50 Marks